

Accelerating Motion: Graphs of Accelerating Motion (e.g., Freefall)

Synopsis: You may have seen that a freefalling object does not fall at a constant speed (see, for example, the ticker tape dot spacing - the dot distance increases). Instead, it speeds up as it falls, i.e., this is an accelerated motion. The distance vs. time graph is not a straight line with a constant slope (remember, the slope in distance vs. time is speed) – it's curved (here, it curves up)! The curved distance vs. time graph encodes increasing velocity with time. The velocity increases linearly with time – the velocity vs. time plot is a straight line with a constant slope. (How is this plot different from constant speed motion velocity vs. time?) What is the slope in velocity vs. time? It is acceleration, and in this case, a constant equal to the acceleration due to gravity, denoted with a symbol, g , which is a magnitude and never carries a sign. With upward defined as positive, $a = -g = -9.8 \text{ m/s}^2$ (what is the meaning of the negative sign?) In this Note, we learn how to extract velocity and acceleration information from a curved distance vs. time graph.

Performance Expectations: Recognize that the distance vs. time graph is curved in accelerated motion, and its slope is *not* constant. The velocity or the slope of the distance vs. time linearly varies with time. The velocity vs. time graph is a straight line with a constant slope equal to the constant acceleration.

Preamble

Answer the following questions:

- 1) Why does the distance vs. time graph curve up?

Speed is distance per _____. So, if speed increases, then the distance covered for the same time interval also _____ with time. This results in the graph _____ up.

- 2) In constant-speed motion, the speed vs. time graph is a flat line at a constant value (quite obviously). How should the speed vs. time graph be different with a constant acceleration motion?

Because the speed increases at a constant rate, the speed vs. time graph lin_____ly incre_____s with time. This means the graph of speed vs. time has a cons_____ slope, equaling the value of ac_____.

- 3) Is the direction of motion the same or opposite to the pull of gravity? _____
- 4) Does the object speed up or slow down? _____
- 5) If a downward direction is DEFINED to be negative, explain in ONE sentence why a negative acceleration, $a = -g$, NECESSARILY means the object speeds up downward.

Because the directions of v _____ and a _____ are the same (downward), the object s _____ u _____.

Sketch motion graphs

Constant speed motion (acceleration = 0)	Accelerated motion (acceleration $\neq 0$, constant)	
Steady speed	Speeding up	Slowing down
Distance vs. Time		
Speed vs. Time		
Acceleration vs. Time		

In Calculus Notation

$v = dx/dt$ and $a = dv/dt$. For freefall, acceleration is constant ($a = -g$), so $dv/dt = -g$. Integrate once to find $v(t)$, then integrate again to find $x(t)$:

$$v(t) = v_0 - gt \quad y(t) = y_0 + v_0 t - \frac{1}{2}gt^2$$

Show the two integration steps below: